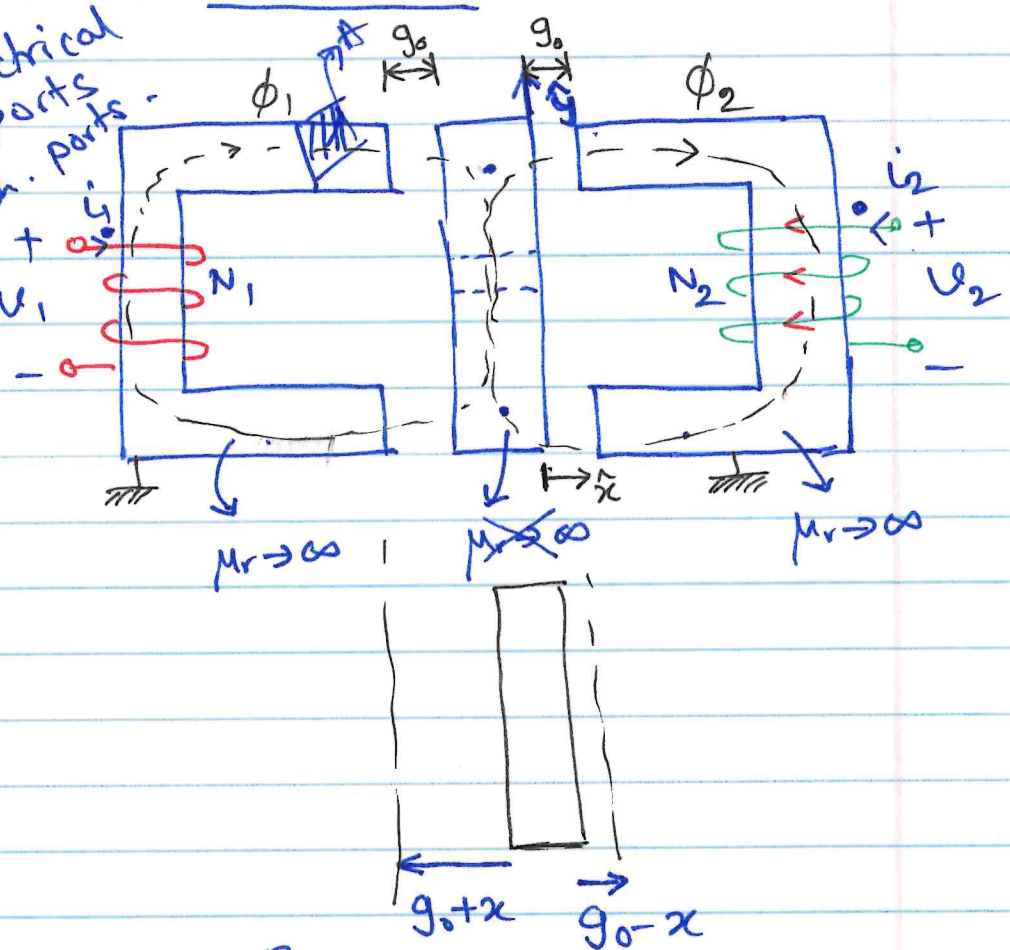
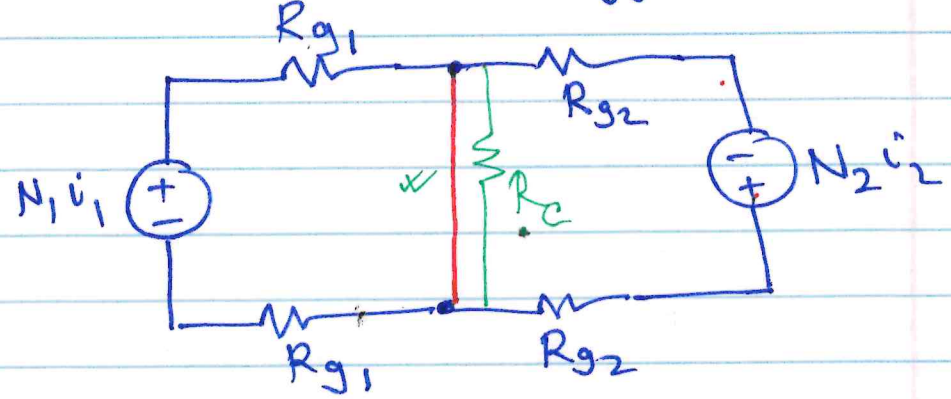
Lecture 16

2 electrical ports.
1 mech. ports.





$(\mu_r \to \infty)$ $\quad \lambda_1 = \boxed{\frac{N_1^2}{2R_{g_1}}} i_1 \qquad \lambda_2 = \frac{N_2^2}{2R_{g_2}} \cdot i_2$

~~$(\mu_r \to \infty)$~~

$$\boxed{\begin{aligned} \lambda_1 &= L_{11} i_1 + M i_2 \\ \lambda_2 &= M i_1 + L_{22} i_2 \end{aligned}}$$

$$\lambda(x, i) = L(x)\, i$$

$$\lambda_1(x, i_1, i_2) = L_{11}(x)\, i_1 + M(x)\, i_2$$

$$\lambda_2(x, i_1, i_2) = M(x)\, i_1 + L_{22}(x)\, i_2$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} L_{11} & M \\ M & L_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{bmatrix} = \begin{bmatrix} \quad \end{bmatrix}_{n \times n} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_n \end{bmatrix}$$

energy method

$$dW_m(\lambda_1, \lambda_2, x) = i_1 d\lambda_1 + i_2 d\lambda_2 - f_e dx$$

CO energy method.

$$dW_m'(i_1, i_2, x) = \lambda_1 di_1 + \lambda_2 di_2 + f_e dx$$

$$W_m'(0,0,0) \rightarrow W_m'(i_1, i_2, x) \rightarrow f_e = \left.\frac{\partial W_m'}{\partial x}\right|_{i_1, i_2 \text{ const}}$$

$$W_m(0,0,0) \rightarrow W_m(\lambda_1, \lambda_2, x) \rightarrow f_e = -\left.\frac{\partial W_m}{\partial x}\right|_{\lambda_1, \lambda_2 \text{ const}}$$

$$W_m'(0,0,0) \rightarrow W_m'(x,0,0) = 0$$

$$W_m'(x, 0 \rightarrow i_1, 0) = \int_0^{i_1} \lambda_1' \, di_1'$$

$$= \int_0^{i_1} L_{11} i_1' \, di_1'$$

$$= \frac{1}{2} L_{11} i_1^2$$

$$W_m'(x, i_1, 0 \rightarrow i_2) = \frac{1}{2} L_{11} i_1^2 + \int_0^{i_2} \lambda_2 \, di_2'$$

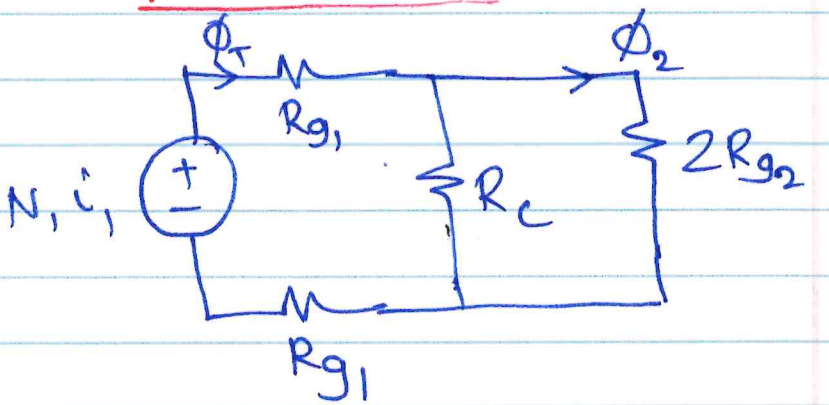
$$= \frac{1}{2} L_{11} i_1^2 + \int_0^{i_2} (M i_1 + L_{22} i_2') \, di_2'$$

$$= \frac{1}{2} L_{11} i_1^2 + M i_1 i_2 + \frac{1}{2} L_{22} i_2^2$$

$$f_e = \left. \frac{\partial W_m'}{\partial x} \right|_{\substack{i_1 = \text{const} \\ i_2 = \text{const}}} = i_1^2 \frac{1}{2} \frac{\partial L_{11}}{\partial x} + i_1 i_2 \frac{\partial M}{\partial x} + i_2^2 \cdot \frac{1}{2} \frac{\partial L_{22}}{\partial x}$$

M : ?

$$\boxed{\lambda_2 = M i_1} \qquad i_2 = 0$$



$$\phi_T = \frac{N_1 i_1}{2R_{g_1} + (2R_{g_2} \| R_c)}$$

$$\phi_2 = \frac{R_c}{R_c + 2R_g} \cdot \phi_T .$$

$$M = N_2 \cdot \frac{R_c}{R_c + 2R_g} N_1 \frac{1}{2R_{g_1} + (2R_{g_2} \| R_c)}$$

$$\boxed{\begin{aligned} \lambda_1 &= L_{11}^{(x)} i_1 + M^{(x)} i_2 \\ \lambda_2 &= M^{(x)} i_1 + L_{22}^{(x)} i_2 \end{aligned}}$$

$$L_{11} = \frac{N_1^2}{2R_{g_1}} \qquad M = 0$$

$$L_{22} = \frac{N_2^2}{2R_{g_2}}$$

$$R_{g_1} = \frac{g_0 + x}{\mu_0 A}$$

$$R_{g_2} = \frac{g_0 - x}{\mu_0 A}$$

$$L_{11} = \frac{N_1^2}{2} \frac{\mu_0 A}{g_0 + x}$$

$$L_{22} = \frac{N_2^2}{2} \frac{\mu_0 A}{g_0 - x}$$

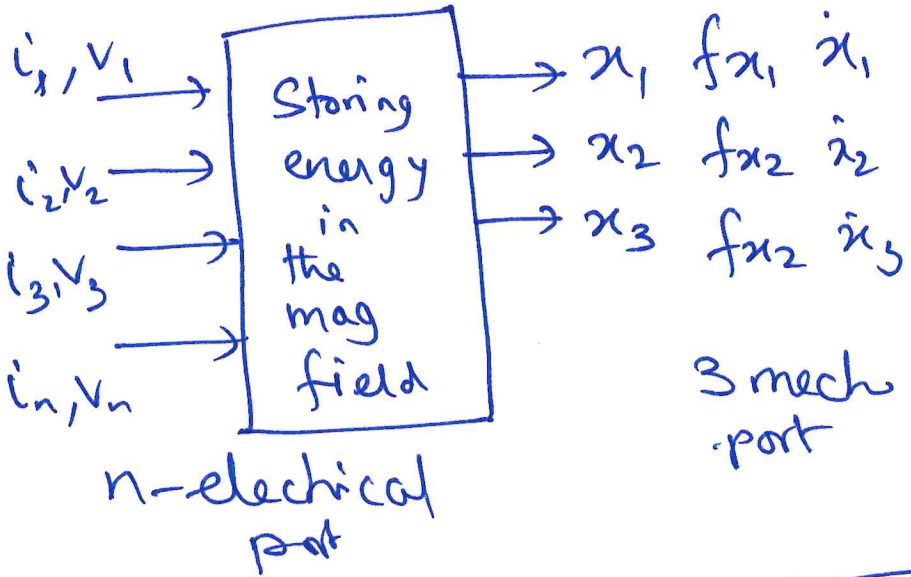
$$f_e = \frac{i_1^2}{2} \frac{\partial L_{11}}{\partial x} + \frac{i_2^2}{2} \frac{\partial L_{22}}{\partial x}$$

$$= \frac{N_1^2 i_1^2 \mu_0 A}{2} \frac{\partial}{\partial x} \cdot \frac{1}{(g_0 + x)} + \frac{N_2^2 i_2^2 \mu_0 A}{2} \frac{\partial}{\partial x} \left(\frac{1}{g_0 - x} \right)$$

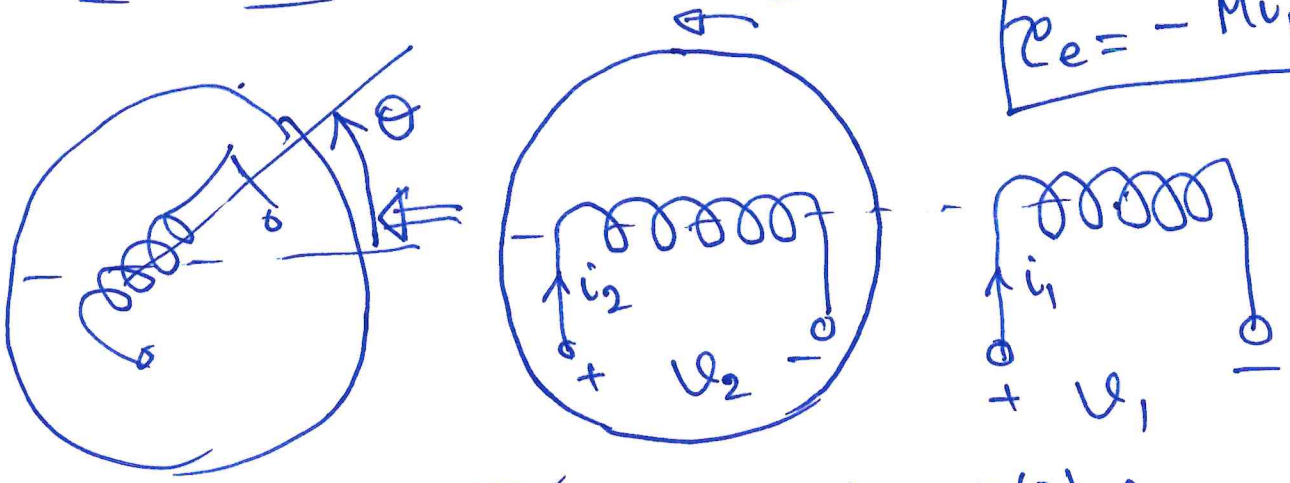
$$= \frac{\mu_0 A}{2} \left[\frac{-N_1^2 i_1^2}{(g_0 + x)^2} + \frac{N_2^2 i_2^2}{(g_0 - x)^2} \right]$$

$$= \frac{\mu_0 A}{2} \left[\frac{N_2^2 i_2^2}{(g_0 - x)^2} - \frac{N_1^2 i_1^2}{(g_0 + x)^2} \right]$$

$$d(W_m(\lambda_1, \lambda_2, x, y)) = i_1 d\lambda_1 + i_2 d\lambda_2 - f_{ex} dx - f_{ey} dy$$



$$T_e = -M i_1 i_2 \sin\theta$$



$$\lambda_1 = L_{11} i_1 + M\cos\theta\, i_2$$
$$\lambda_2 = M\cos\theta\, i_1 + L_{22} i_2$$

$$\lambda_1 = L_{11} i_1 + M(\theta) i_2$$
$$\lambda_2 = M(\theta) i_1 + L_{22} i_2$$

~~Q.~~

$$W_m(\lambda_1, \lambda_2, \theta)$$

$$W_m'(i_1, i_2, \theta)$$

$$\tau_e = \left.\frac{\partial W_m'}{\partial \theta}\right|_{i_1, i_2 \text{ const}} = \frac{i_1^2}{2}\frac{\partial L_{11}}{\partial \theta} + \frac{i_2^2}{2}\frac{\partial L_{22}}{\partial \theta} + i_1 i_2 \frac{\partial M}{\partial \theta}$$

$$\tau_e = \left[\frac{\partial}{\partial \theta} \cdot M\cos\theta\right] i_1 i_2$$

$$= -M i_1 i_2 \sin\theta$$